

Radians



1 Calculate the following trigonometric ratios to two decimal places:

a $\sin \frac{35\pi}{16}$

b $\cos 6.87$

c $\sin 7.26$

d $\tan 7.26$

e $\tan \left(-\frac{3\pi}{7} \right)$

f $\cos \frac{2\pi}{3}$

g $\sin \left(-\frac{4\pi}{3} \right)$

h $\cos \frac{4\pi}{5}$

i $\sin \left(-\frac{4\pi}{5} \right)$

Exact values

2 Consider the following diagram:

a Find the length of side h .

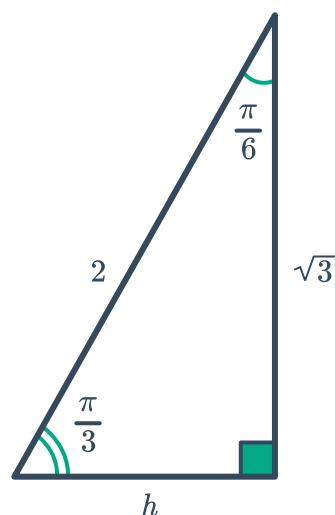
b Hence, state the exact value of:

i $\sin \frac{\pi}{3}$

ii $\sin \frac{\pi}{6}$

iii $\tan \frac{\pi}{3}$

iv $\cos \frac{\pi}{3}$



3 Consider the following diagram:

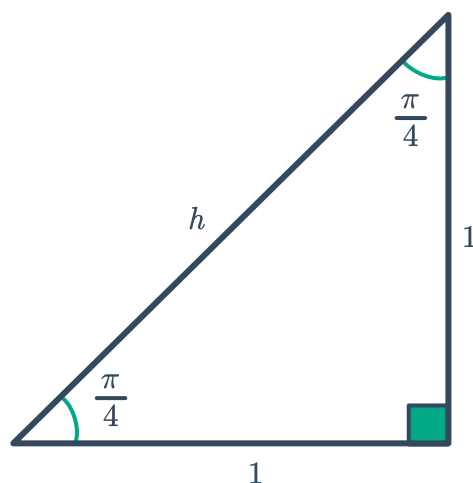
a Find the length of the hypotenuse, h .

b Hence, state the exact value of:

i $\sin \frac{\pi}{4}$

ii $\cos \frac{\pi}{4}$

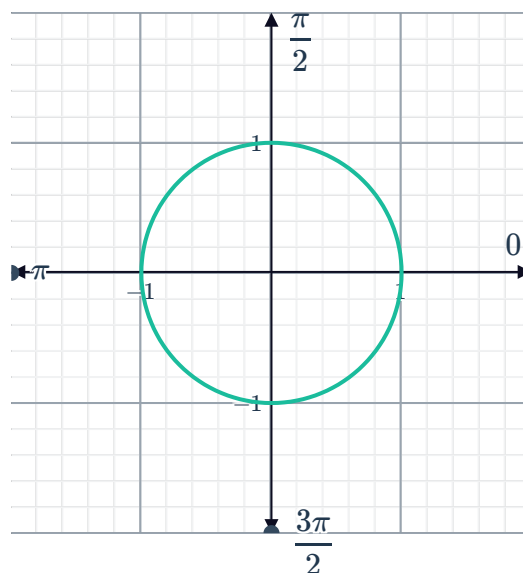
iii $\tan \frac{\pi}{4}$



- 4 Consider the unit circle diagram and state the exact value of the following trigonometric ratios:



- a $\sin \frac{\pi}{2}$ b $\cos \frac{3\pi}{2}$
 c $\tan \pi$ d $\cos 0$
 e $\sin(-2\pi)$

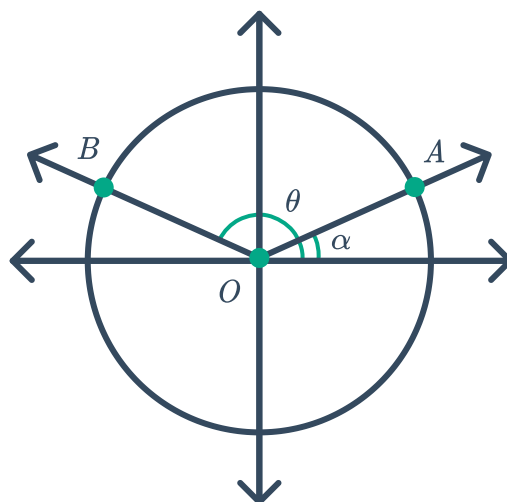


- 5 Find the exact value of the following:

- a $\sin \frac{\pi}{3} + \cos \frac{\pi}{3}$ b $\sin \frac{\pi}{6} \cos \frac{\pi}{4}$
 c $\frac{\sin \frac{\pi}{3}}{\cos \frac{\pi}{6}}$ d $\sin \frac{\pi}{4} \cos \frac{\pi}{6} + \tan \frac{\pi}{4}$
 e $\sin^2 \left(\frac{\pi}{6} \right) - \cos^2 \left(\frac{\pi}{3} \right)$ f $2 \sin^2 \left(\frac{\pi}{2} \right) + 3 \cos^2 \left(\frac{\pi}{2} \right)$

Exact values from reference angles

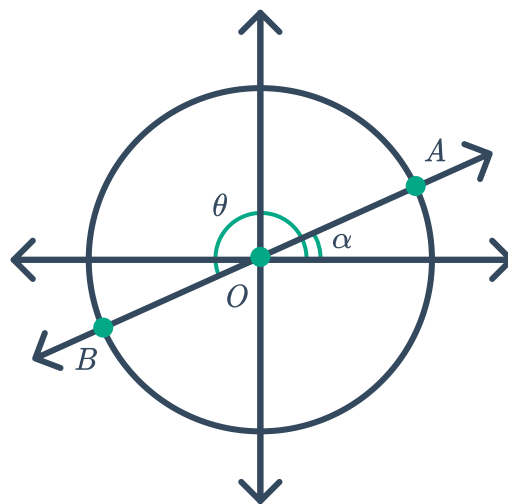
- 6 Consider the unit circle shown, where points A and B have the same y -coordinates.
 Suppose that $\theta = \frac{10\pi}{11}$. State the size of the reference angle, α .



- 7 Consider the unit circle shown, where the line through A and B passes through the origin, O .

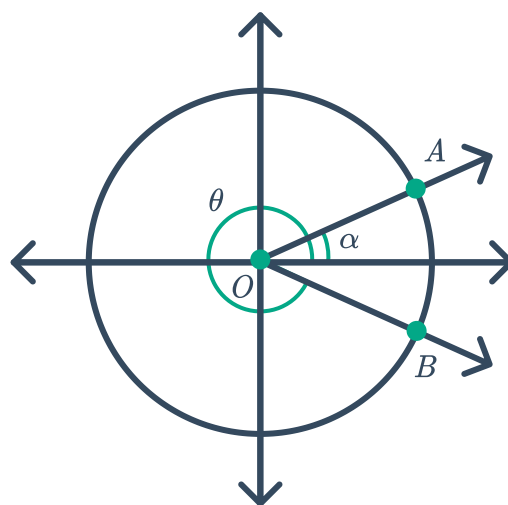


Suppose that $\theta = \frac{8\pi}{7}$. State the size of the reference angle, α .



- 8 Consider the unit circle shown, where the points A and B have the same x -coordinate.

Suppose that $\theta = \frac{9\pi}{5}$. State the size of the reference angle, α .



- 9 Find the exact value of the following:

a $\sin \frac{5\pi}{6}$

b $\tan \frac{3\pi}{4}$

c $\sin \frac{7\pi}{6}$

d $\cos \frac{7\pi}{6}$

e $\sin \frac{5\pi}{3}$

f $\cos \frac{5\pi}{3}$

g $\cos 4\pi$

h $\tan 9\pi$

i $\sin \frac{5\pi}{2}$

j $\cos \frac{7\pi}{2}$

k $\cos \frac{3\pi}{4}$

l $\sin \frac{5\pi}{4}$

m $\cos \frac{5\pi}{4}$

n $\tan \frac{5\pi}{4}$

o $\tan \frac{7\pi}{6}$

p $\tan \frac{11\pi}{6}$

- 10 Find the exact value of the following:

a $\sin \left(-\frac{17\pi}{6} \right)$

b $\cos \left(-\frac{17\pi}{6} \right)$

c $\cos \left(-\frac{4\pi}{3} \right)$

d $\tan \left(-\frac{17\pi}{6} \right)$

11 Find the exact value of the following:



a
$$\frac{\left(\sin \frac{2\pi}{3}\right) \left(\cos \frac{2\pi}{3}\right) \left(\tan \frac{3\pi}{4}\right)}{\tan \left(-\frac{\pi}{4}\right)}$$

b
$$\frac{\sin \frac{2\pi}{3} + \cos \frac{5\pi}{6} - \tan \frac{7\pi}{4}}{\cos \frac{4\pi}{3}}$$

Trigonometric functions

12 Consider the equation $y = \sin x$.

a Complete the table with values in exact form:

x	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{3\pi}{2}$	$\frac{11\pi}{6}$	2π
$\sin x$									

b Sketch a graph for $y = \sin x$ on the domain $-2\pi \leq 0 \leq 2\pi$.

c State the value of $\sin(-2\pi)$.

d State the sign of $\sin\left(\frac{-\pi}{12}\right)$.

e State the sign of $\sin \frac{13\pi}{12}$.

f Which quadrant of a unit circle does an angle with measure $\frac{13\pi}{12}$ lie in?

13 Consider the equation $y = \cos x$.

a Complete the table with values in exact form:

x	0	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	2π
$\cos x$									

b Sketch a graph for $y = \cos x$ on the domain $-2\pi \leq 0 \leq 2\pi$.

c State the value of $\cos \pi$.

d State the sign of $\cos\left(\frac{-\pi}{4}\right)$.

e State the sign of $\cos \frac{11\pi}{6}$.

f Which quadrant of a unit circle does an angle with measure $\frac{11\pi}{6}$ lie in?

14 Consider the equation $y = \tan x$.



a Complete the table with values in exact form:

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\tan x$									

b Sketch the graph of $y = \tan x$ on the domain $-2\pi \leq 0 \leq 2\pi$.

c Graph the line $y = 1$ on the same coordinate plane.

d Hence, state the exact solutions to the equation $\tan x = 1$ over this domain.

e State the value of $\tan(-2\pi)$.

f State the sign of $\tan\left(\frac{-\pi}{6}\right)$.

g State the sign of $\tan \frac{9\pi}{5}$.

h Which quadrant of a unit circle does an angle with measure $\frac{9\pi}{5}$ lie in?

15 Consider the function $y = 5 \sin x$.

a Sketch the graph of the given function over the domain $[-2\pi, 2\pi]$.

b Graph the line $y = x$ on the same coordinate plane.

c Hence, state the number of solutions to the equation $5 \sin x = x$.

16 Consider the graph of $y = -\tan x$ and the plotted points A, B, C, D and E shown:
At which point is the graph of $y = -\tan x$ equal to zero?

